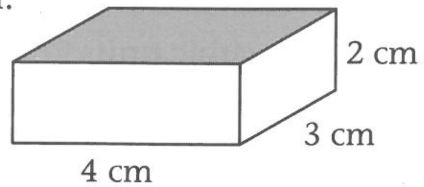
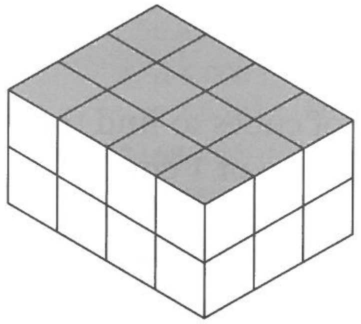
4.2 Volume of a rectangular prism A

A small box is the shape of a rectangular prism.
It is 4 cm long, 3 cm wide and 2 cm deep.
Binh wants to calculate the volume.



EXAMPLE

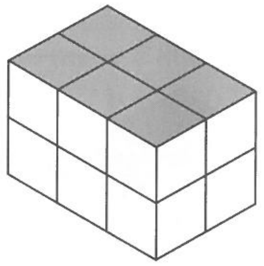
This rectangular prism is made using 1 cm cubes.
Each cube has a volume of 1 cm^3.
The prism has two layers.
Each layer has 12 cubes (3 rows of 4 cubes).



Volume of this prism
= 12 × 2
= 24 cm^3

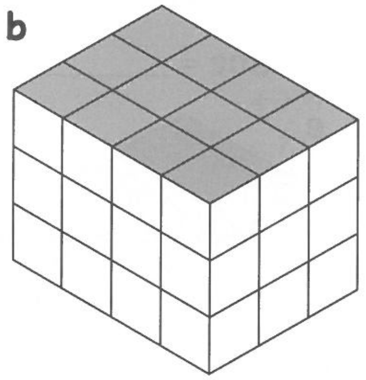
1 These rectangular prisms are made using 1 cm cubes.
Count the number of cubes in one layer. Then multiply by the number of layers. This will give the volume of the prism.

a



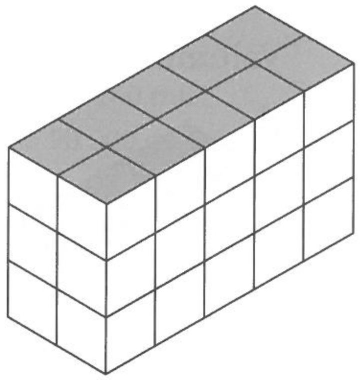
Volume = ____________

b



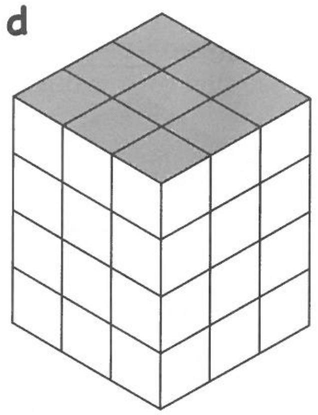
Volume = ____________

c



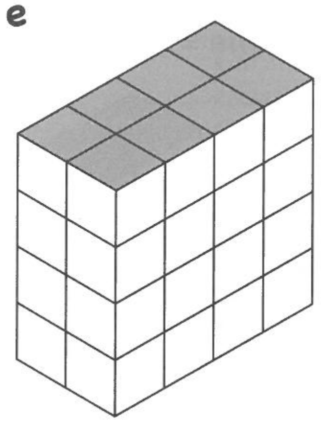
Volume = ____________

d



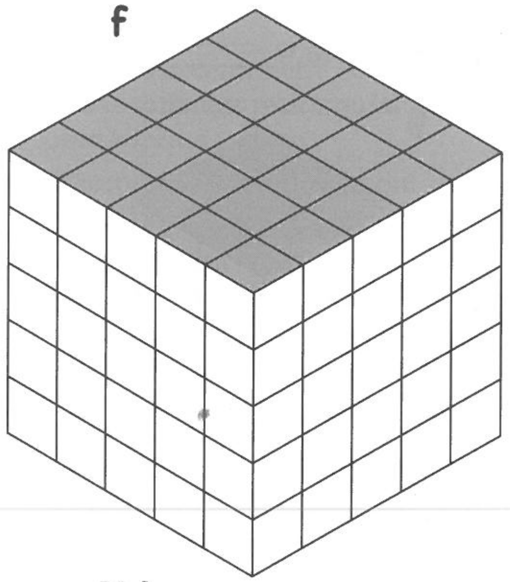
Volume = ____________

e



Volume = ____________

f

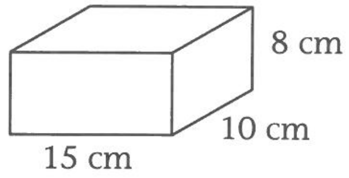


Volume = ____________

4.3 Volume of a rectangular prism B

To find the volume of a rectangular prism, first find the area of the base.
(Multiply length × width.)
Then multiply by the height.
Volume = length × width × height

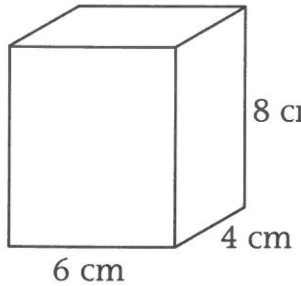
EXAMPLE



The volume of this rectangular prism is:
15 × 10 × 8
= 1 200 cm^3

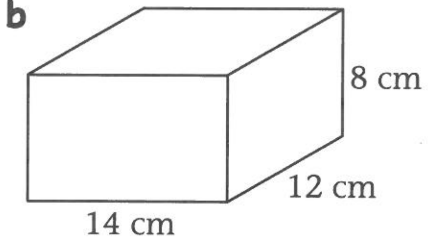
1 Use the rule **volume = length × width × height** to calculate the volume of each rectangular prism.

a



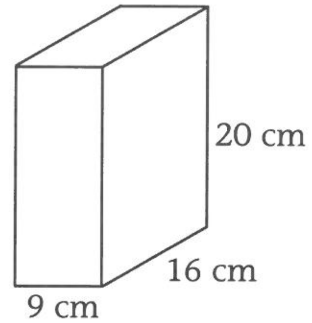
Volume = ____________
= ____________

b



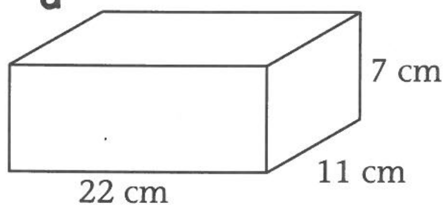
Volume = ____________
= ____________

c



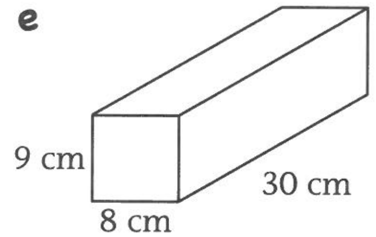
Volume = ____________
= ____________

d



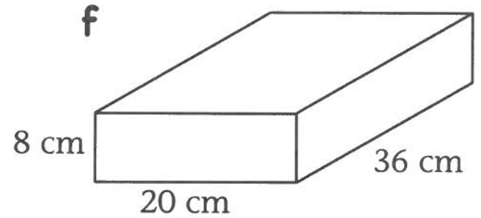
Volume = ____________
= ____________

e



Volume = ____________
= ____________

f



Volume = ____________
= ____________

Answer the following questions in your workbook.

2 A swimming pool is the shape of a rectangular prism. It is 25 metres long, 8 metres wide and 2 metres deep. Calculate its volume.

3 A rectangular water storage tank measures 3.5 m by 1.4 m by 2.8 m. Calculate its volume.

4 The internal dimensions of a refrigerator are: height = 180 cm, width = 74 cm and depth = 60 cm. Calculate its volume.